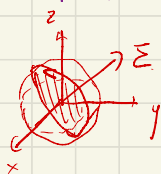


Stokes 公式的应用

(1) $\oint_{\Gamma} y dx + z dy + x dz$, $\Gamma: x^2 + y^2 + z^2 = a^2$, $x + y + z = 0$

在 z 轴上的截距, 用极坐标



取平面的法向量

$$(\cos \alpha, \cos \beta, \cos \gamma) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

$$\begin{aligned} \oint_{\Gamma} y dx + z dy + x dz &= \iint_{\Sigma} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} ds \\ &= \iint_{\Sigma} (-\cos \alpha - \cos \beta - \cos \gamma) ds = -\sqrt{3} \iint_{\Sigma} ds = -\sqrt{3} \pi a^2 \end{aligned}$$

(2) $\oint_{\Gamma} (y-z) dx + (z-xy) dy + (x-y) dz$, $\Gamma: x^2 + y^2 = a^2$, $\frac{x}{a} + \frac{y}{b} = 1$ ($a > 0, b > 0$)

在 xy 平面上的投影



$$\vec{n} = \left(\frac{a}{\sqrt{a^2 + b^2}}, 0, \frac{b}{\sqrt{a^2 + b^2}} \right)$$

$$\begin{aligned} \oint_{\Gamma} &= \iint_{\Sigma} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-z & z-xy & x-y \end{vmatrix} ds \\ &= -2 \iint_{\Sigma} (\cos \alpha + \cos \beta - \cos \gamma) ds \\ &= \frac{-2 \cos \alpha}{\sqrt{a^2 + b^2}} \iint_{\Sigma} ds = \frac{-2 \cos \alpha}{\sqrt{a^2 + b^2}} \iint_{D} \frac{\sqrt{a^2 + b^2}}{a} dx dy = \frac{-2 \cos \alpha}{a} \iint_{D} dx dy = -2 \cos \alpha \cdot \pi ab \end{aligned}$$

(3) $\oint_{\Gamma} 3y dx - xz dy + yz^2 dz$, $\Gamma: x^2 + y^2 = 2z$, $z=2$. z 轴上的截距, 用极坐标

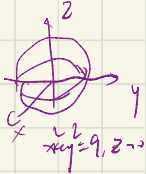


解: $\oint_{\Gamma} 3y dx - xz dy + yz^2 dz$

取平面的法向量

$$\begin{aligned} &= \iint_{\Sigma} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & -xz & yz^2 \end{vmatrix} ds = \iint_{\Sigma} (z^2 + x) dy dz - (z + 2) dx dy \Rightarrow r = 2, S = 4\pi \\ &= \iint_{\Sigma} (4 + x) dy dz - 5 dx dy = 0 - 5 \iint_{\Sigma} dx dy = -5 \times S = -20\pi \end{aligned}$$

4. $\oint_{\Gamma} 2y dx + 3xy dy - z^2 dz$, $\Gamma: x^2 + y^2 + z^2 = 4$, $z=0$ 在 xy 平面上的投影



$$\begin{aligned} &\oint_{\Gamma} 2y dx + 3xy dy - z^2 dz \\ &= \iint_{\Sigma} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & 3xy & -z^2 \end{vmatrix} ds = \iint_{\Sigma} (3-2) dx dy = \iint_{\Sigma} dx dy = 5\pi \end{aligned}$$